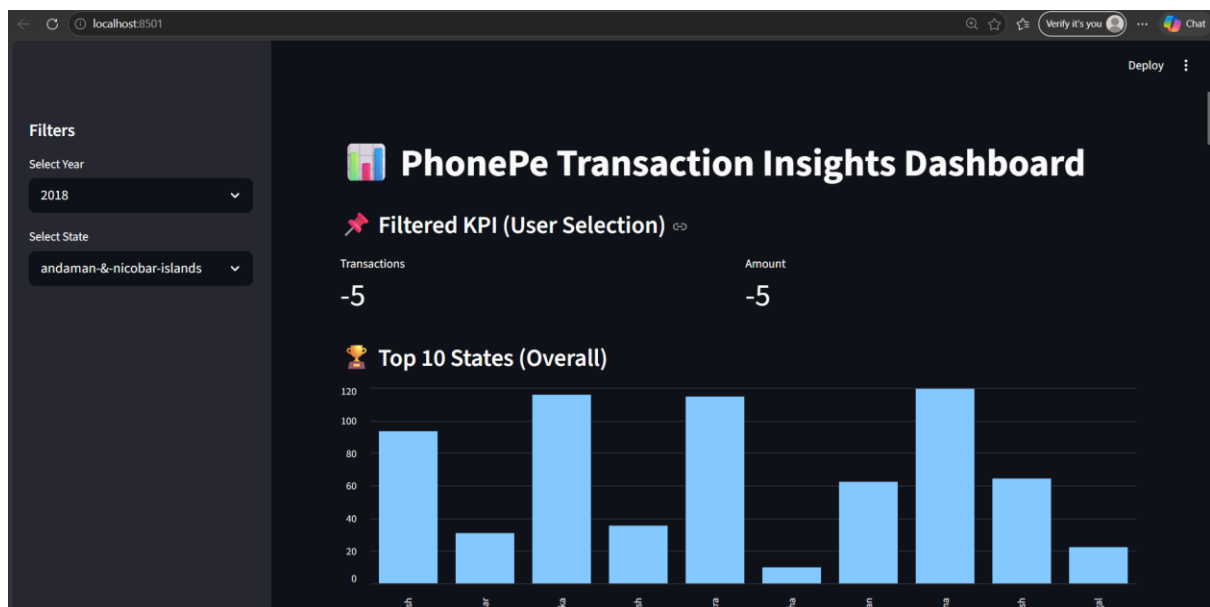


PhonePe Transaction Insights Dashboard – Report

1. Introduction

This project presents a PhonePe Transaction Insights Dashboard built using Python, Streamlit, Pandas, SQLite, Matplotlib, and Seaborn. The goal is to analyze real transaction data and extract meaningful insights such as state-wise performance, user behavior, transaction trends, and insurance usage.

The dashboard provides interactive filters and visualizations to help understand digital payment patterns in India.



2. Dataset Description

The system uses three datasets stored in SQLite:

- **transactions table**
 - year, state, quarter, transaction_type, count, amount
- **users table**
 - brand, count (device usage)
- **insurance table**

- year, amount

These datasets are loaded dynamically using SQL queries inside Streamlit.

3. Data Processing

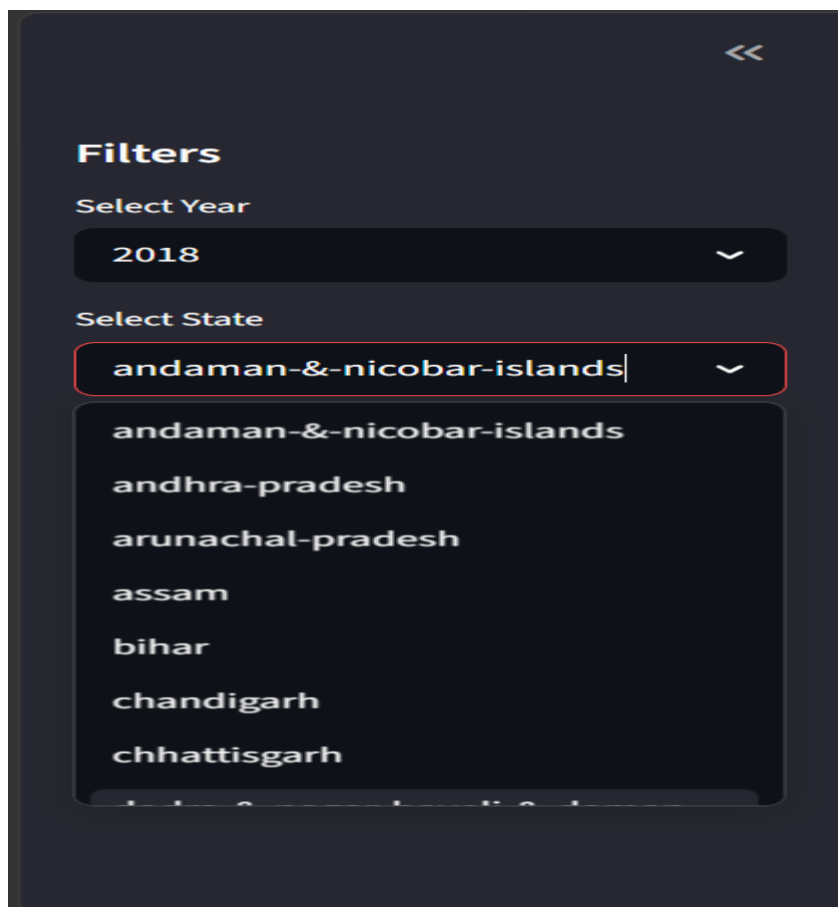
Data is fetched using SQLite and converted into Pandas DataFrames.

Aggregations are performed using `groupby()` for:

- State-wise transaction amount
- Brand-wise user distribution
- Yearly and quarterly trends

Filtering is applied using sidebar selections for:

- Year
- State



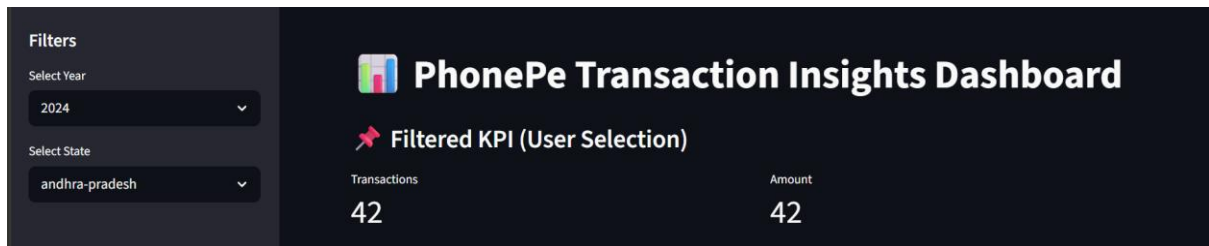
4. Key Performance Indicators (KPI)

This section shows filtered insights based on user selection.

Metrics displayed:

- Total Transactions
- Total Amount

These values dynamically change based on selected year and state.



5. State-wise Transaction Analysis

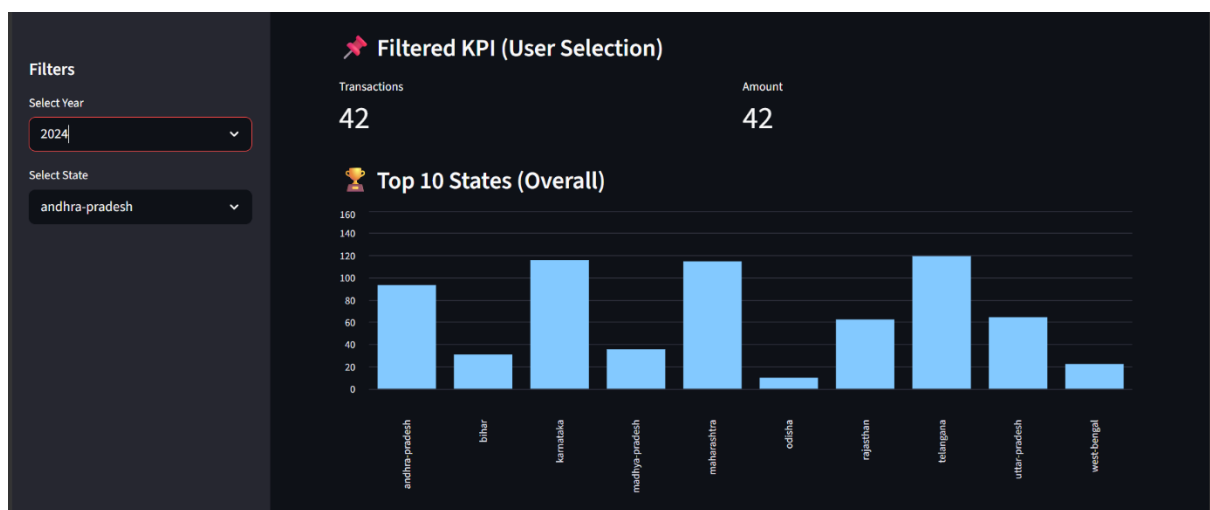
This visualization shows the **top 10 states contributing to transaction volume**.

It is calculated using:

```
df_trans.groupby("state")["amount"].sum()
```

Insight:

- A few states dominate total digital transactions
- Shows regional concentration of digital payments



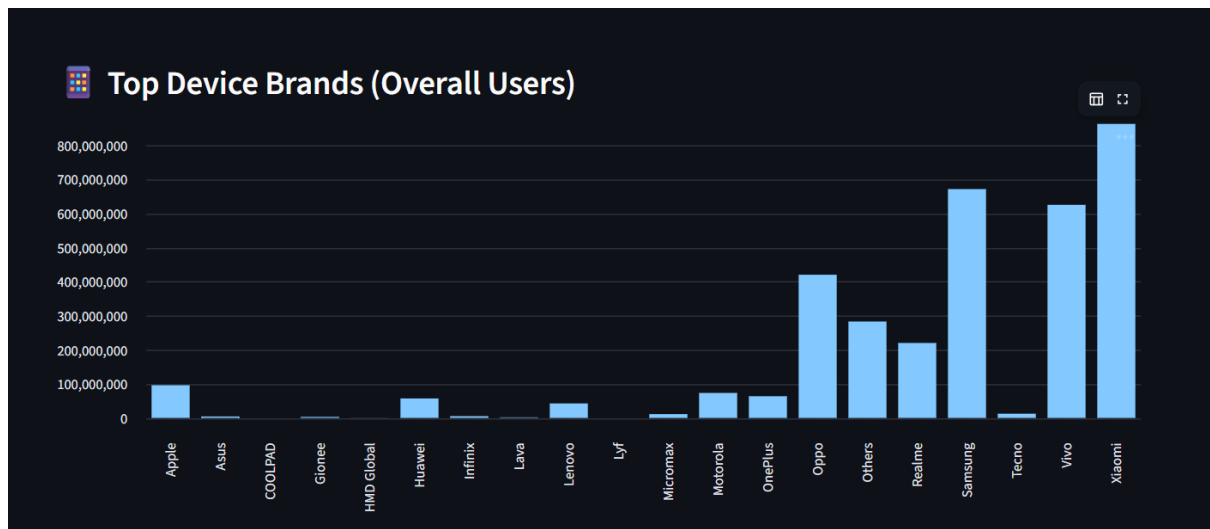
6. Device Brand Analysis (User Behavior)

This section analyzes **mobile device usage among users**.

Brands like Xiaomi, Samsung, Vivo, etc. are ranked based on usage count.

Insight:

- Xiaomi and similar mid-range brands dominate usage
- Indicates affordability-driven smartphone adoption

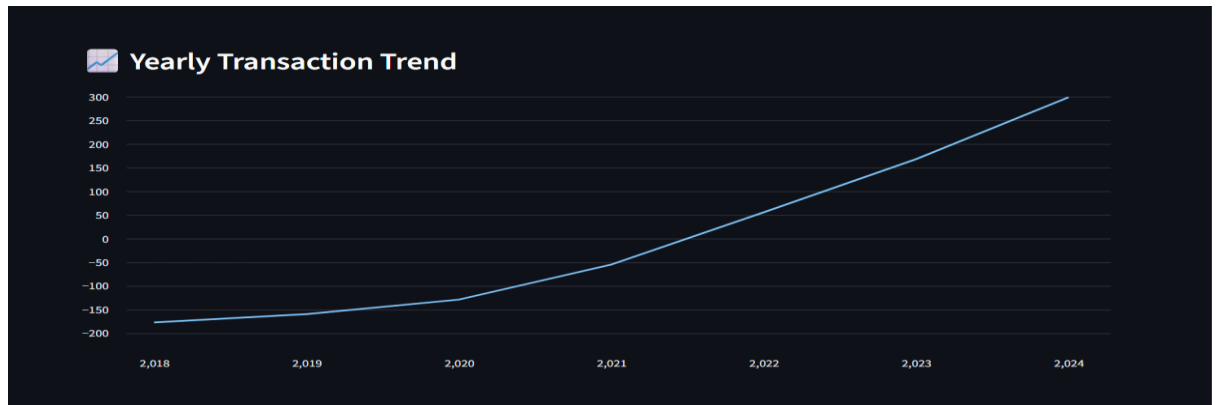


7. Yearly Transaction Trend

This shows how transaction volume changes over years.

Insight:

- Continuous growth indicates increasing digital adoption
- Helps track financial digitization trends

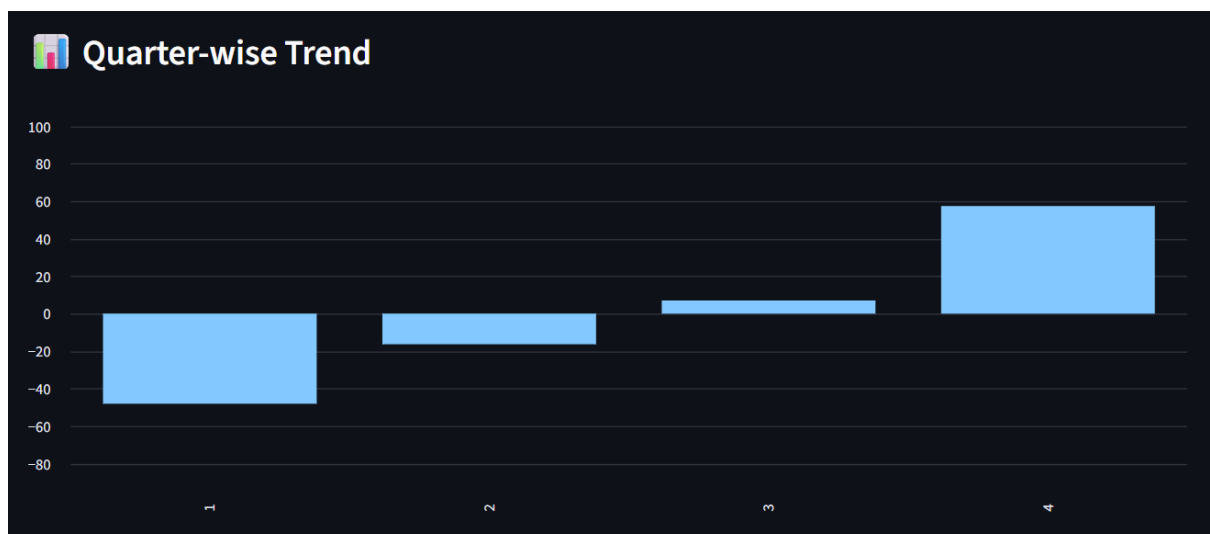


8. Quarter-wise Analysis

This visualization breaks transactions into quarters (Q1–Q4).

Insight:

- Seasonal variations in transaction volume
- Useful for business forecasting



9. Transaction Type Distribution

This chart shows distribution of transaction categories like:

- Recharge
- Peer-to-peer transfer
- Merchant payments (depending on dataset)

Insight:

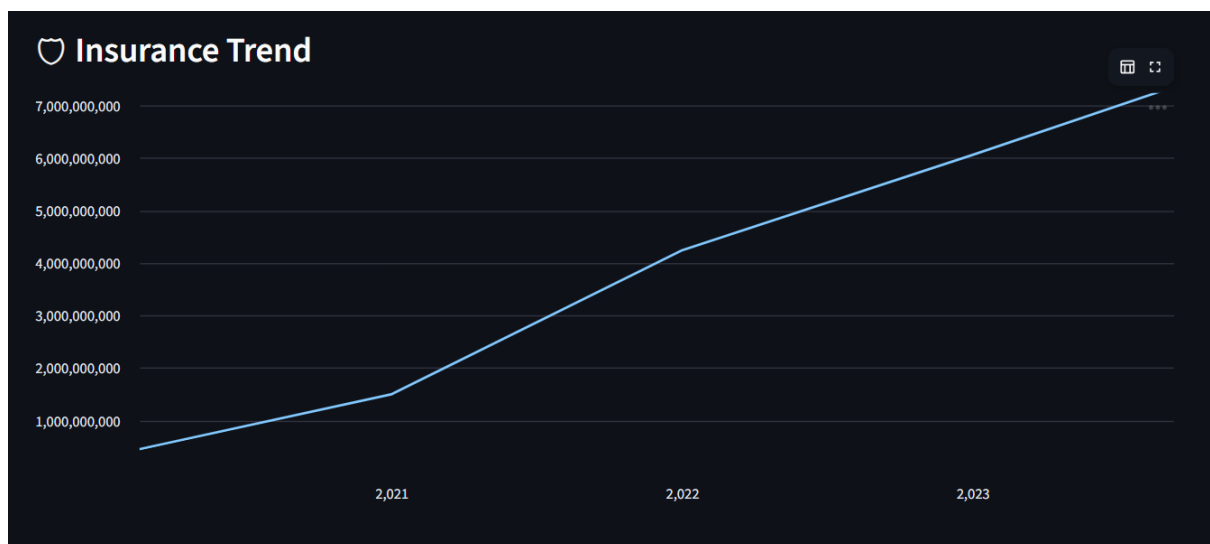
- Identifies most commonly used payment type
- Helps understand user behavior patterns

10. Insurance Analysis

This section shows insurance transaction trends over years.

Insight:

- Growth indicates increasing digital insurance adoption
- Helps financial institutions track adoption patterns



11. Conclusion

The dashboard successfully visualizes PhonePe transaction data and provides insights into:

- State-wise transaction dominance
- User device preferences
- Transaction growth trends
- Insurance adoption patterns

This system demonstrates the use of:

- Data analytics
- SQL database integration
- Interactive dashboards using Streamlit